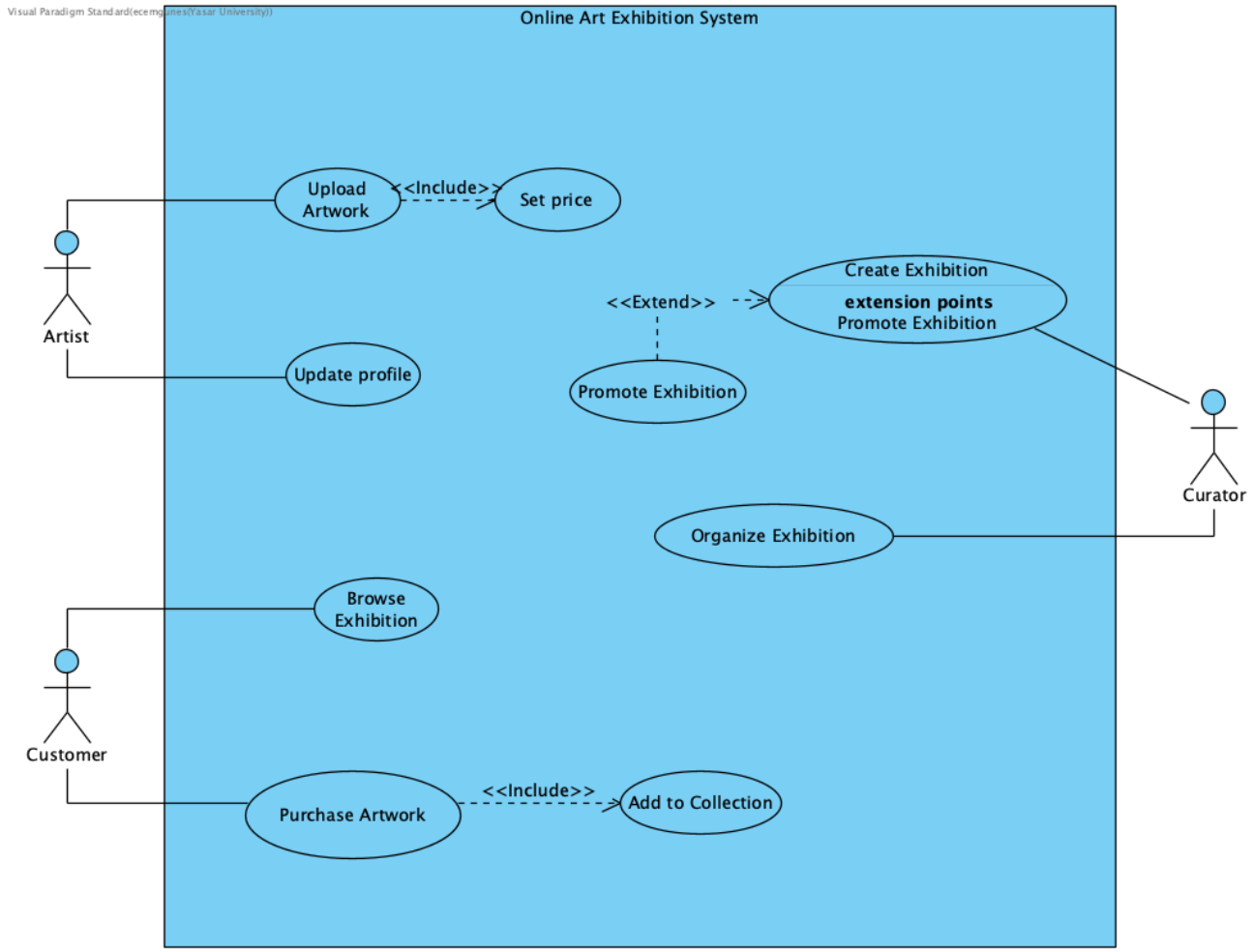
A.Part 1. SYSTEM ANALSYSIS REPORT

1. PROJECT PROPOSAL FORM

Name Surname:
Ecem GÜNEŞ
Project Codename:
ARTSYS
Project Title:
Online Art Exhibition System
Project Brief Summary:
The Online Art Exhibition System addresses the challenge of limited visibility and accessibility for artists who wish to share and sell their artwork. Traditional galleries have space, cost, and geographic limitations that hinder exposure, especially for emerging artists.
This platform provides a digital alternative where artists can showcase and monetize their work online. The system enables curators to create thematic exhibitions, and customers to easily explore and purchase artworks. This not only enhances access for art-lovers but also provides a structured and interactive environment for digital art engagement.
Project Dependencies: Users:
The system will include three well-defined user roles:
Artist: Can create an account, upload artworks with titles and descriptions, set prices, and manage their personal portfolio.
Curator: Organizes and manages virtual exhibitions by selecting artworks based on a theme. They also earn a 1% commission per artwork included in their exhibition.
Customer: Can register, browse exhibitions, view artwork details, purchase pieces, and store purchased items in a personal collection.
Project Dependencies: Data storage:
The system will store data related to ;
-Users' information: Account information of artists, curators and customers. (Username, password, e-mail, profile info.)
-Art-works: title, description, price, image, upload date
-Exhibition: name, theme, participating artists
-Transaction: purchase details, order status, payment info.
Data storage way ;
-Database for store user accounts, artwork data, exhibition details, and transaction records.
-Cloud storage for artwork images
Project Dependencies: Graphical User Interface:
The system will be implemented as a Desktop Application using PyQt5. The GUI will include separate interface views tailored to each user role:
Artist Interface: Artwork upload panel, portfolio management screen, and earnings summary.
Curator Interface: Exhibition creation and management dashboard, with options to set themes and promote exhibitions.
Customer Interface: Art browsing gallery, detailed artwork view, cart, and personal collection section.
The interface will prioritize visual clarity, user-friendliness, and intuitive navigation to ensure a smooth experience for all user types.

2. USE CASE DIAGRAM



3. USE CASE DESCRIPTION

3.1 UPLOAD ARTWORK

Use Case Description:
This use case allows an artist to upload a new piece of artwork to the platform. The artist provides details such as the title, description, price, and uploads an image.
Use Case Name:
Upload Artwork
Actor:
Artist
Triggers:
Artist uploads a new artwork to the platform, including image and metadata (title,description,price).
Preconditions:
1. 1.Artist must be logged into the system.(Artist is authenticated and is on the dashboard.)
2. 2.Artist must be setting the price of art pieces. (include)
3. Artwork image must meet the required file format and size limits.
Main Flow (Basic Scenario):
1. Artist selects “Upload Artwork” option.
2. System displays the upload form.
3. Artist enters artwork title and description.
4. Artist uploads an image.
5. System validates the file and displays fields to set price. **(INCLUDE: Set Price)**
6. Artist completes the form and submits.
7. System saves the artwork and image.
8. System shows confirmation.
Alternative Flows:
1. If required fields are missing or invalid, system notifies artist.
2. If the image format is unsupported (e.g., PDF), system shows an error and prompts for a valid image file.
3. If artist cancels the process, system returns to dashboard without saving.
Postconditions:
1. Artwork is stored in the database and associated with the artist.
2. Artwork is ready to be part of exhibitions.

3.1.1 Set Price <<include>>

Use Case Description:
This use case describes how an artist sets the price for an artwork during the upload process. It ensures the artwork has a defined value before it becomes available for exhibitions or purchase.
Use Case Name:
Set Price
Actor:
Artist
Triggers:
Artist is uploading a new artwork and reaches the pricing section. *(INCLUDE from Upload Artwork)*
Preconditions:
1. 1.Artist is logged in.
2. 2.Artwork upload form is active.
Main Flow (Basic Scenario):
1. 1.System prompts artist to enter a price.
2. 2.Artist enters a numeric value.
3. 3.System validates the price format.
4. 4.System saves the price data temporarily in the upload session.
Alternative Flows:
1. 1.If the price is not a valid number, system shows an error.
2. 2.If artist leaves the price blank, system does not allow submission.
3. 3.If artist cancels, system clears unsaved data.
Postconditions:
Price is associated with the artwork and saved along with it.

3.2 CREATE EXHIBITION

Use Case Description:
This use case describes how a curator creates a new exhibition by selecting artworks and setting a theme. Optionally, the curator may promote the exhibition afterward.
Use Case Name:
Create Exhibition
Actor:
Curator
Triggers:
Curator selects the "Create Exhibition" option.
Preconditions:
1.Curator must be logged in.
2.Artworks are available in the system.
Main Flow (Basic Scenario):

1. Curator selects "Create Exhibition" option.
2. System displays exhibition form.
3. Curator enters exhibition name and theme.
4. System shows available artworks.
5. Curator selects artworks.
6. Curator confirms creation.
7. System saves and publishes the exhibition.
8. System calculates and transfers 1% commission from each artwork listed in the exhibition to the curator's earnings.(include:)
9. (EXTEND: Promote Exhibition — optional after creation.)
10. Show confirmation.

Alternative Flows:

1. If required fields are missing, system shows error.
2. If no artworks are selected, system prevents submission.
3. If curator cancels, system returns to dashboard without saving.
4. Upon publishing the exhibition, the system calculates and transfers 1% commission from each artwork listed in the exhibition to the curator's earnings.

Postconditions:

1. A new exhibition is available on the platform.
2. It can be optionally promoted.

3.2.1 Promote Exhibition <<extend>>

Use Case Description:
This use case describes how a curator can promote an exhibition after it has been created, to increase visibility among art-lovers.
Use Case Name:
Promote Exhibition
Actor:
Curator
Triggers:
Curator selects the “Promote” option after creating an exhibition. *(EXTEND from Create Exhibition)*
Preconditions:
1.Curator is logged in.
2.An existing exhibition has been successfully created.
Main Flow (Basic Scenario):
1.Curator selects “Promote Exhibition.”
2.System shows promotion options (e.g., featuring, banners, highlighting).
3.Curator chooses a promotion method.
4.System applies promotion and updates visibility settings.
5.(EXTEND: Promote Exhibition — optional after creation.)
Alternative Flows:
1.If promotion settings are not selected properly, system shows error.
2.If curator cancels the process, system returns to the exhibition page.
Postconditions:
Exhibition gains increased visibility on the platform.

3.2.2 Receive Commission <<include>>

Use Case Description:
This use case describes how the system automatically calculates and transfers a 1% commission from each artwork included in a newly created exhibition to the curator’s account.
Use Case Name:
Receive Commission
Actor:
Curator
Triggers:
Included during the execution of the “Create Exhibition” use case.
Preconditions:
1.A valid exhibition creation process is in progress.
2.Artworks are selected and assigned to the exhibition.
Main Flow (Basic Scenario):
1.System identifies the artworks included in the exhibition.
2.System calculates 1% of the price of each artwork.
3.System adds the total commission amount to the curator’s earnings or account.
4.System logs the commission transaction for record-keeping.
Alternative Flows:
1.If any artwork lacks pricing information, the system skips that item and notifies the curator.
2.If curator’s profile is invalid or restricted, system prevents commission from being added and logs an error.

Postconditions:
The curator's account reflects the commission income.
A record of commission transaction is stored by the system.

3.2 PURCHASE ARTWORK

Use Case Name:
Purchase Artwork
Actor:
Customer
Use Case Description:
This use case describes how a customer purchases an artwork. Upon successful purchase, the artwork is automatically added to the customer's collection.
Triggers:
Customer selects "Purchase" from the artwork details page.
Preconditions:

1. Customer is logged in.
2. Artwork is available and not sold.

Main Flow (Basic Scenario):
1.Customer selects an artwork.
2.System displays details and "Purchase" button.
3.Customer proceeds to payment.
4.System processes the payment.
5.System confirms success and updates transaction.
6.Artwork is added to customer's collection. **(INCLUDE: Add to Collection)**
7.System shows purchase confirmation.

Alternative Flows:
1.If the customer is not registered, the system redirects to registration and blocks the purchase attempt until registration is complete.
2.If payment fails, system notifies customer.
3.If artwork is already sold, purchase is blocked.

4.If the customer cancels the payment process, the system returns to the artwork detail page and the transaction is not recorded.

Postconditions:
1.Transaction is recorded.
2.Artwork is added to personal collection.

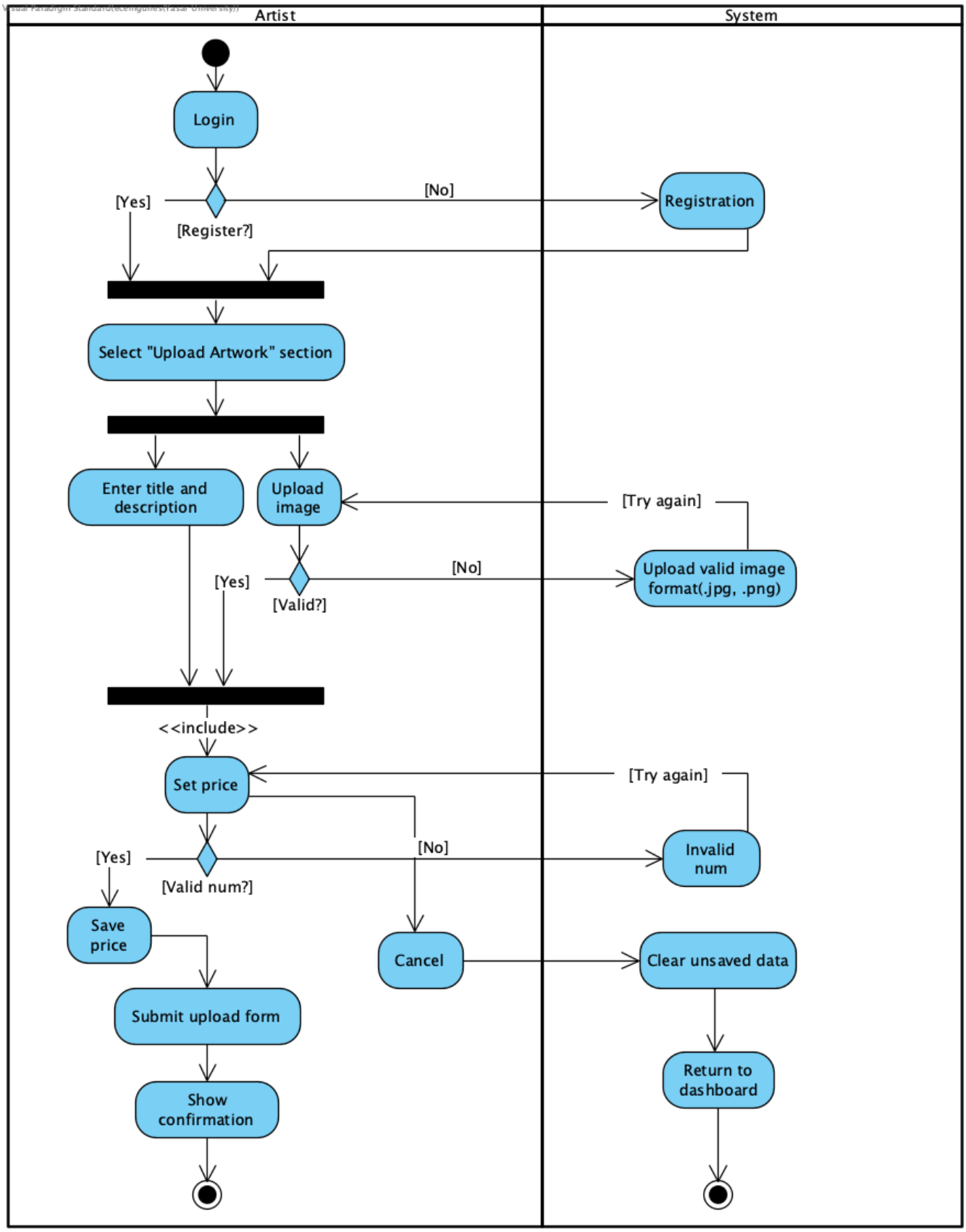
3.1 Add to Collection <<include>>

Use Case Name:
Add to Collection
Actor:
Customer
Use Case Description:
This use case describes how a purchased artwork is added to the customer's personal collection, making it accessible for future viewing or management.
Triggers:
A purchase is successfully completed. *(INCLUDE from Purchase Artwork)*
Preconditions:
1.Customer is logged in.
2.Artwork has been purchased successfully.

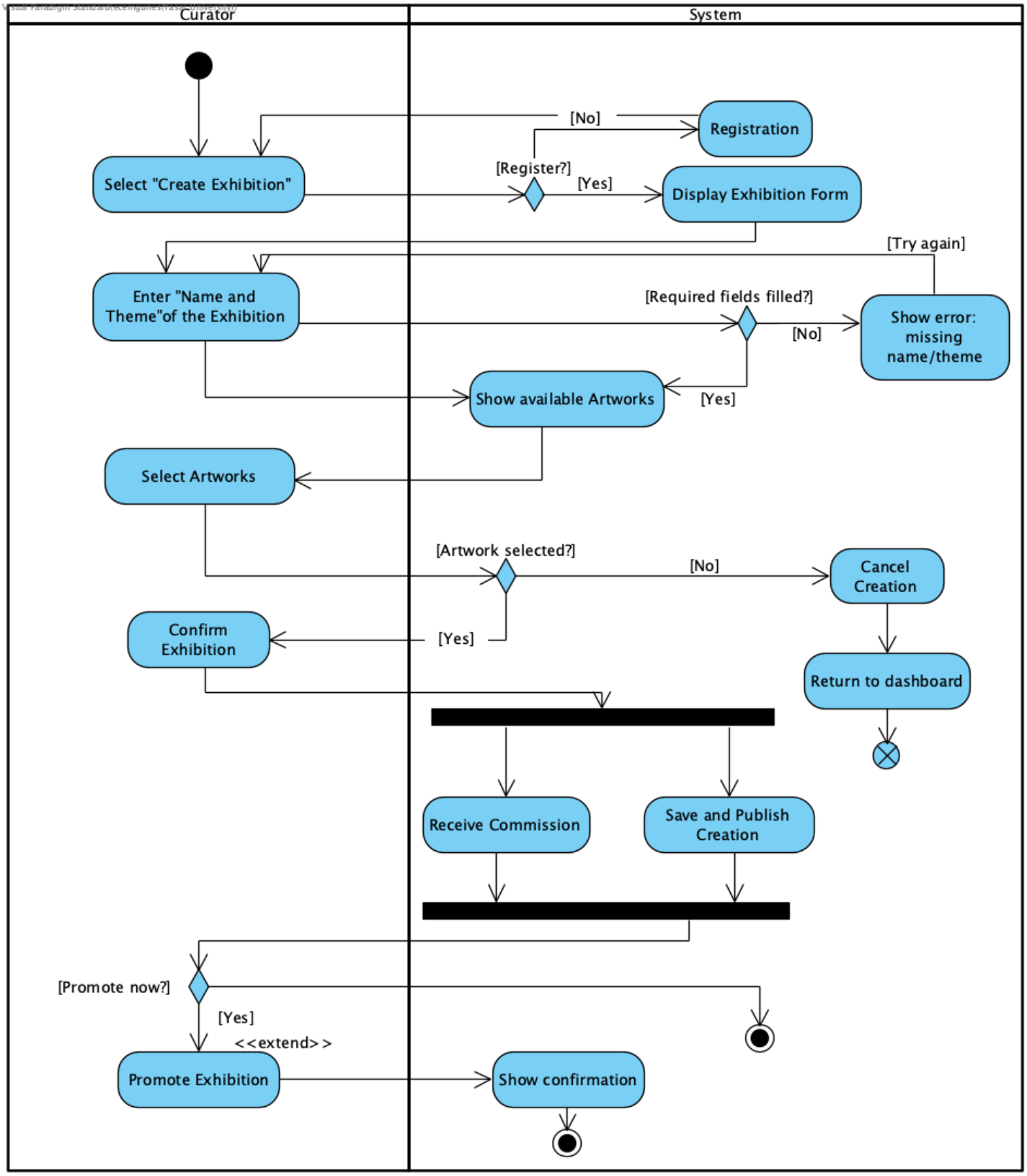
Main Flow (Basic Scenario):
1.System detects a completed purchase.
2.System adds the artwork to the customer's personal collection area.
3.Customer is notified that the artwork has been added.
Alternative Flows:
1.If there's a system error, the addition is delayed and retried.
2.If customer's collection storage is full (if such a limit exists), system notifies.
Postconditions:
Artwork appears in the customer's collection section.

4. ACTIVITY DIAGRAM

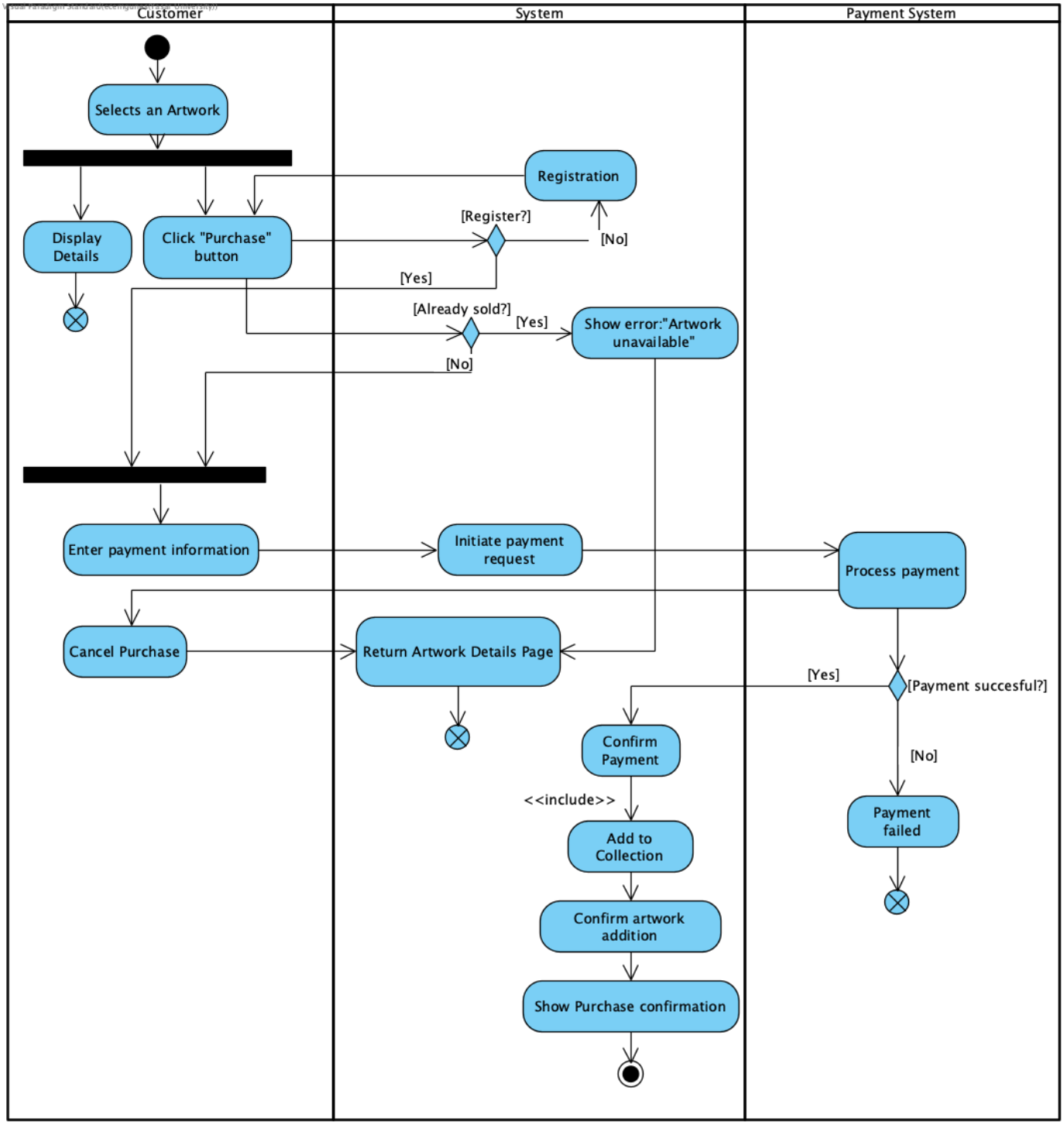
4.1 Upload Artwork


Artist
System
Login
[Yes]
[Register?]
[No]
Registration
Select "Upload Artwork" section
Enter title and description
Upload image
[Try again]
[Yes]
[Valid?]
[No]
Upload valid image format(.jpg, .png)
<<include>>
Set price
[Try again]
[Yes]
[Valid num?]
[No]
Invalid num
Save price
Cancel
Clear unsaved data
Submit upload form
Return to dashboard
Show confirmation

4.2 Create Exhibition

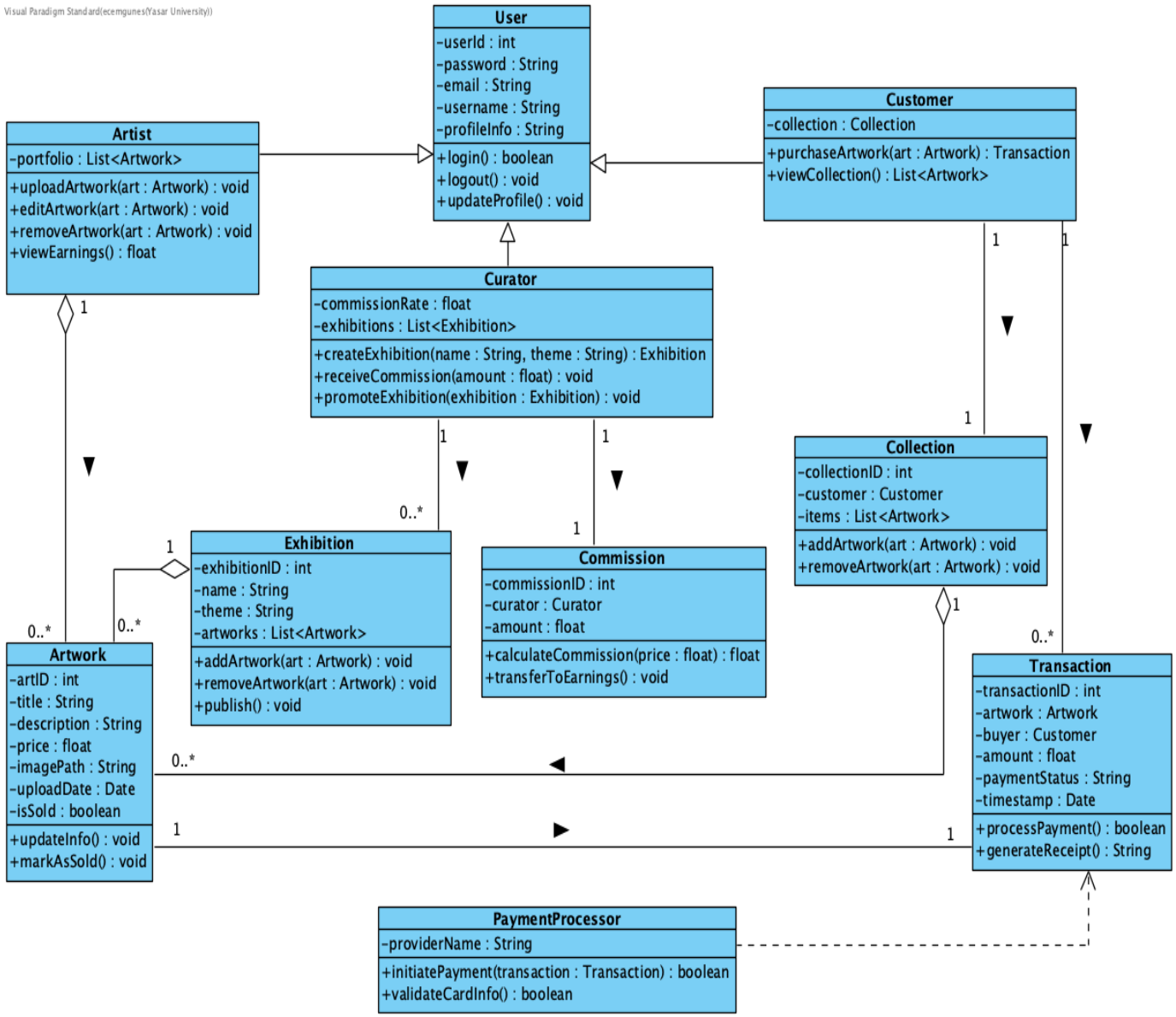

Curator
System
Select "Create Exhibition"
[Register?]
[No]
[Yes]
Registration
Display Exhibition Form
Enter "Name and Theme"of the Exhibition
[Required fields filled?]
[No]
[Yes]
[Try again]
Show error: missing name/theme
Show available Artworks
Select Artworks
[Artwork selected?]
[No]
[Yes]
Cancel Creation
Confirm Exhibition
Return to dashboard
Receive Commission
Save and Publish Creation
[Promote now?]
[Yes]
<<extend>>
Promote Exhibition
Show confirmation

4.3 Purchase Artwork



B.Part 2. SYSTEM DESIGN REPORT

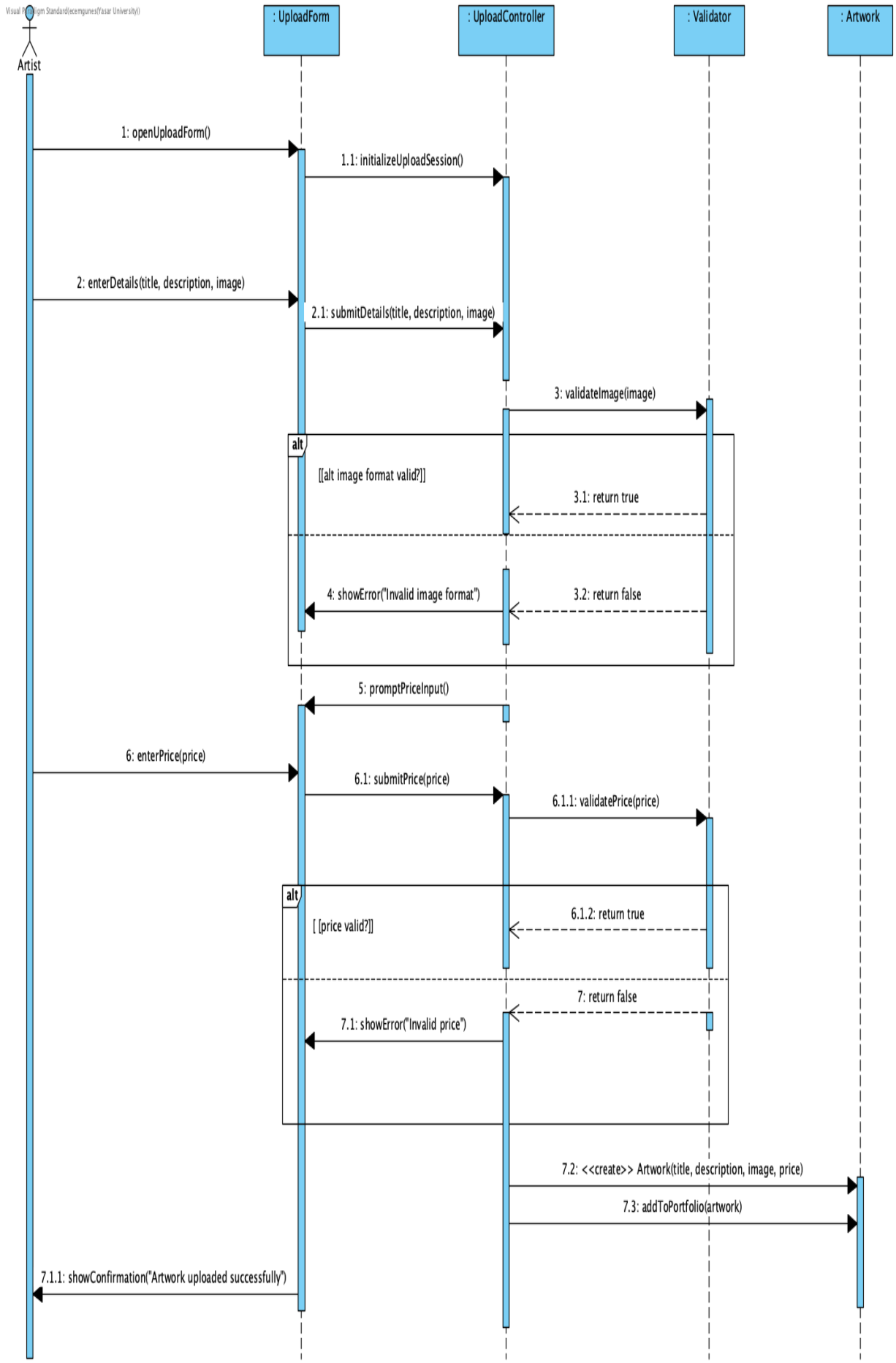
1. DESIGN CLASS DIAGRAM



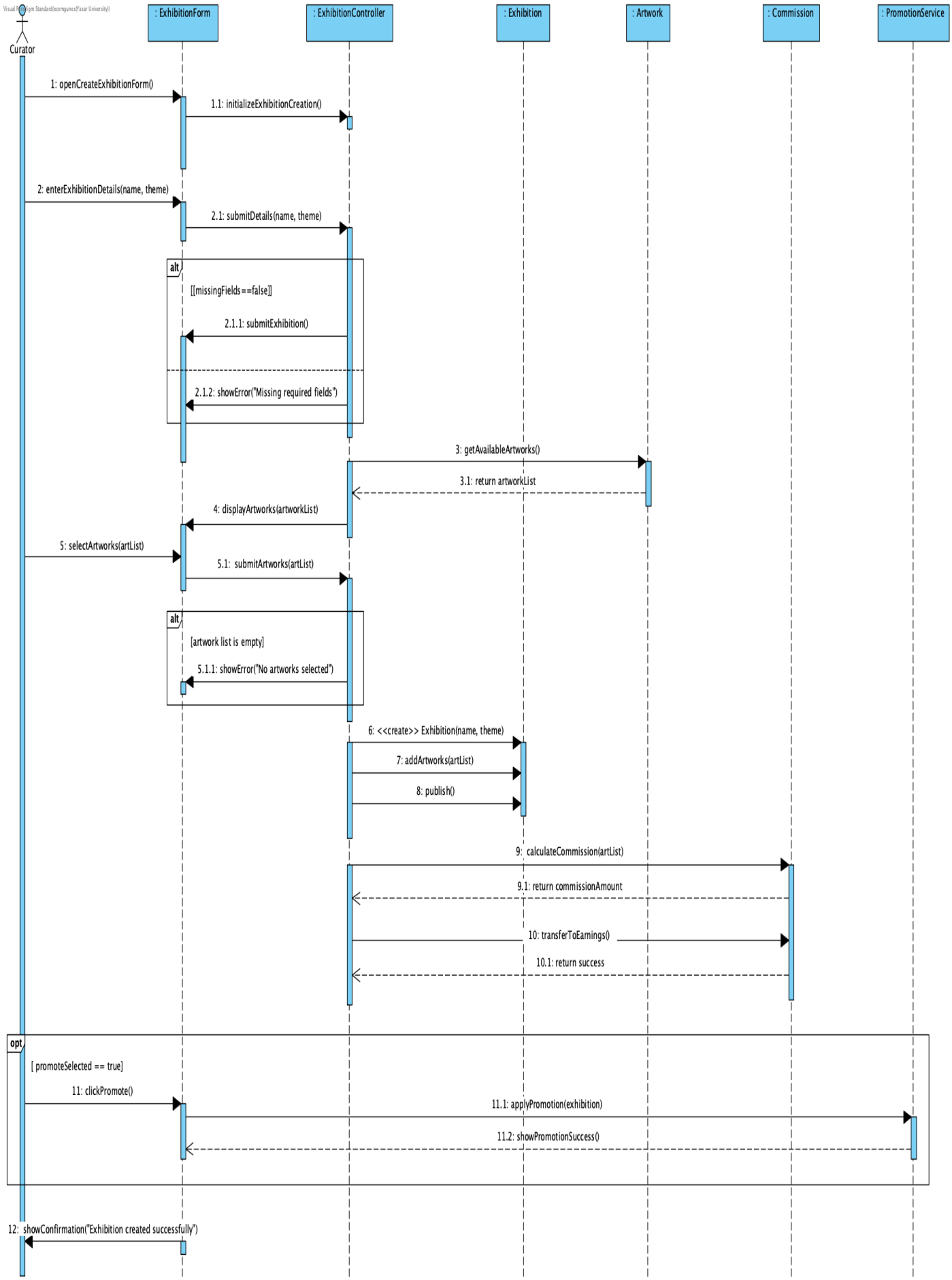
The methods login(), logout(), and updateProfile() are defined in the User superclass and may be overridden in its subclasses (Artist, Curator, and Customer) to provide polymorphic behavior. These methods are not explicitly repeated in the diagram for each subclass, but in implementation, they can be customized to reflect role-specific functionality.

2. SEQUENCE DIAGRAM

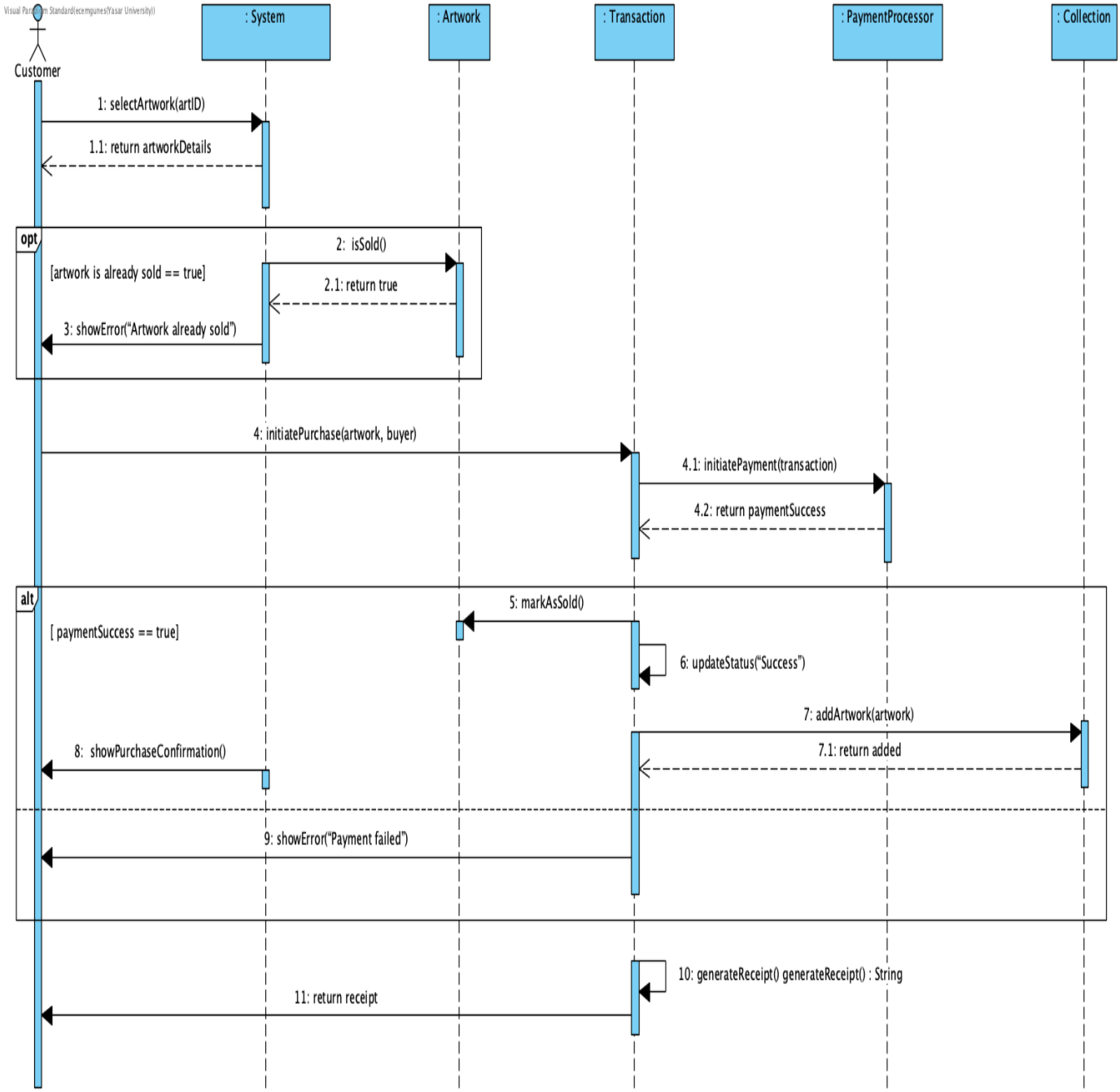
2.1 Upload Artwork


Artist
: UploadForm
: UploadController
: Validator
: Artwork
1: openUploadForm()
1.1: initializeUploadSession()
2: enterDetails(title, description, image)
2.1: submitDetails(title, description, image)
3: validateImage(image)
alt
[[alt image format valid?]]
3.1: return true
4: showError("Invalid image format")
3.2: return false
5: promptPriceInput()
6: enterPrice(price)
6.1: submitPrice(price)
6.1.1: validatePrice(price)
alt
[[price valid?]]
6.1.2: return true
7: return false
7.1: showError("Invalid price")
7.2: <<create>> Artwork(title, description, image, price)
7.3: addToPortfolio(artwork)
7.1.1: showConfirmation("Artwork uploaded successfully")

2.2 Create Exhibition


Visual Paradigm Standard(ecemgunes(Yasar University))
Curator
: ExhibitionForm
: ExhibitionController
: Exhibition
: Artwork
: Commission
: PromotionService
1: openCreateExhibitionForm()
1.1: initializeExhibitionCreation()
2: enterExhibitionDetails(name, theme)
2.1: submitDetails(name, theme)
alt
[[missingFields==false]]
2.1.1: submitExhibition()
2.1.2: showError("Missing required fields")
3: getAvailableArtworks()
3.1: return artworkList
4: displayArtworks(artworkList)
5: selectArtworks(artList)
5.1: submitArtworks(artList)
alt
[artwork list is empty]
5.1.1: showError("No artworks selected")
6: <<create>> Exhibition(name, theme)
7: addArtworks(artList)
8: publish()
9: calculateCommission(artList)
9.1: return commissionAmount
10: transferToEarnings()
10.1: return success
opt
[promoteSelected == true]
11: clickPromote()
11.1: applyPromotion(exhibition)
11.2: showPromotionSuccess()
12: showConfirmation("Exhibition created successfully")

2.3 Purchase Artwork



3.EXPLANATION

3.1 Upload Artwork

The **Upload Artwork** sequence diagram models the interaction between an artist and the system when uploading a new artwork. Initially, the artist selects the upload option, prompting the system to display the upload form. The artist enters the title, description, and image, which is validated by the system. If the image format is invalid, the system shows an error message (opt block). Otherwise, the system proceeds to prompt the artist to enter a price. The price is then validated; if it is invalid, an error is shown, otherwise the process continues. Once all inputs are valid, the system creates an Artwork object and adds it to the artist's portfolio. Finally, a confirmation message is shown. Additionally, alternative flows are handled: if required fields are missing, or if the artist cancels the process, appropriate messages are displayed and the user is returned to the dashboard (modeled with opt blocks). This diagram includes both the main functional flow and all relevant exception scenarios using UML control fragments.

3.2 Create Exhibition

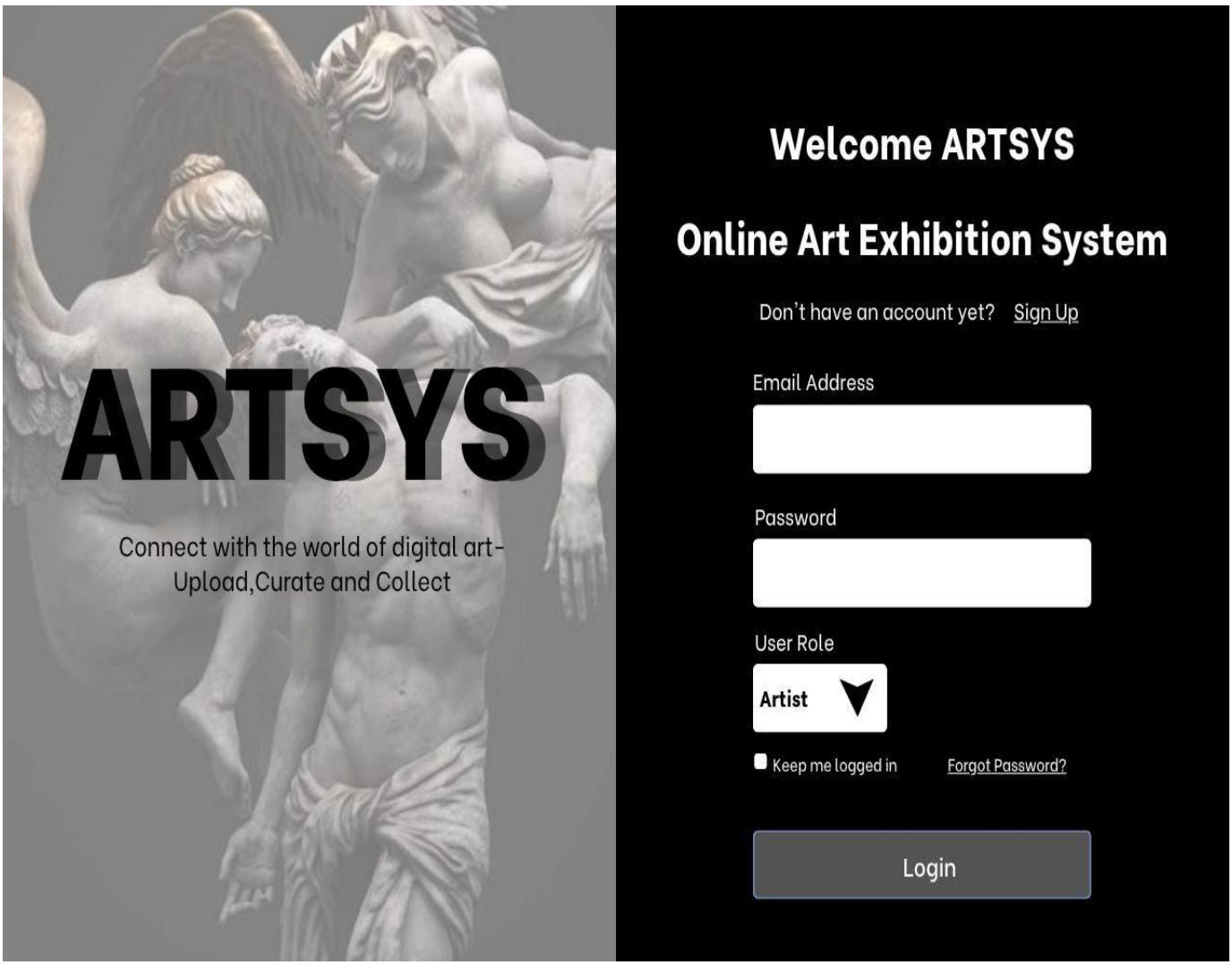
The **Create Exhibition** sequence diagram captures the process in which a curator creates and publishes a new exhibition. The curator begins by selecting the "Create Exhibition" option, leading the system to display the exhibition form. After entering the name and theme, the system shows available artworks. The curator selects artworks and submits the exhibition, triggering the creation of an Exhibition object, the addition of artworks, and the invocation of the publish() method. Subsequently, the system includes an automatic commission calculation and transfer using the Commission class, modeled via the include relationship. Optional promotion is handled with an opt block, where the curator may promote the exhibition using the Promotion Service. Furthermore, the diagram includes key alternative flows using opt fragments—such as when required fields are missing, no artworks are selected, or the curator cancels the process—ensuring a comprehensive representation of both standard and exceptional behaviors.

3.3 Purchase Artwork

The **Purchase Artwork** sequence diagram demonstrates the steps involved when a customer purchases an artwork. The interaction begins when the customer selects an artwork, prompting the system to display its details. If the artwork is already sold, an error is shown (opt block). If available, the customer proceeds with the purchase, and the system initiates payment processing via the PaymentProcessor. Based on the result, the diagram branches with an alt block: if payment is successful, the transaction is marked as complete, the artwork is marked as sold, and it is added to the customer's Collection. A confirmation is then displayed. If the payment fails, the system notifies the customer via an error message. These exception paths, along with the successful transaction flow, are clearly modeled to illustrate both functional success and error-handling scenarios

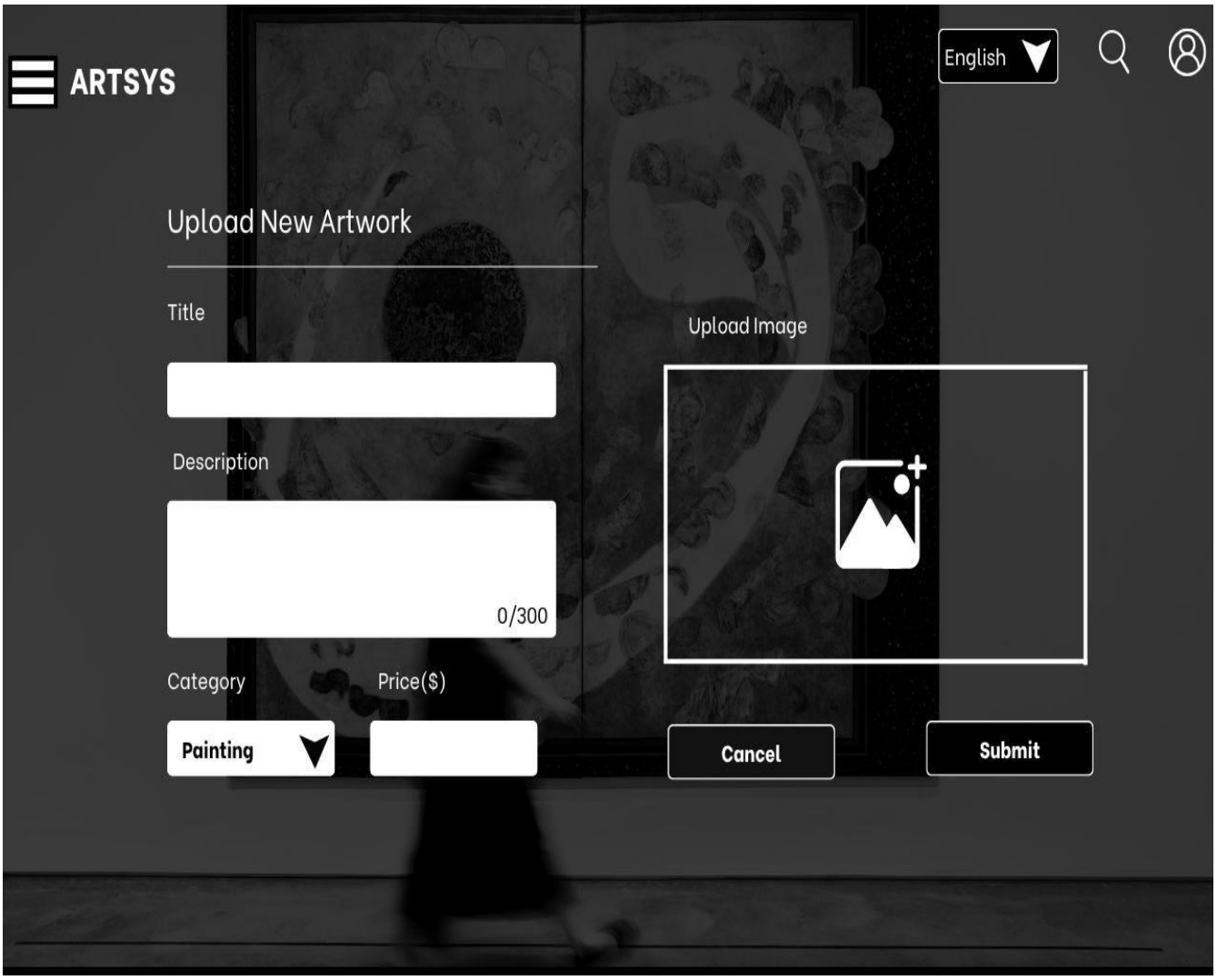
C.Part 3. SCREENSHOTS

1.Login Page

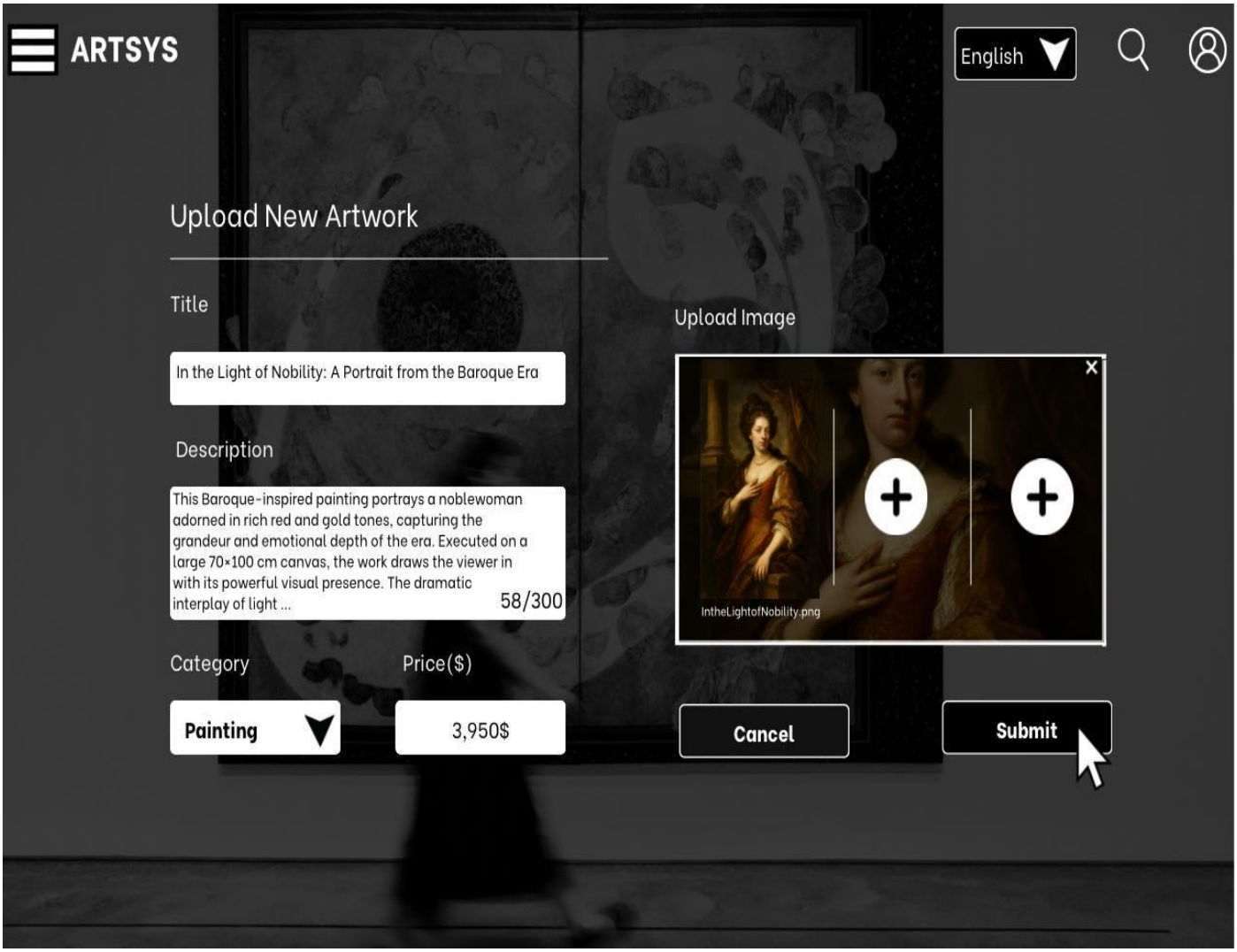


The login screen serves as the entry point to the ARTSYS platform. It allows registered users—Artists, Curators, and Art-Lovers—to securely access the system by providing their email address, password, and selecting their role. The interface features a clean, user-friendly design with high visual contrast and clear form fields. A welcome message introduces the system's purpose, emphasizing connectivity and engagement with digital art. Additional functionalities such as "Keep me logged in", "Forgot Password?" and a link to the sign-up page ensure ease of use and account recovery support.

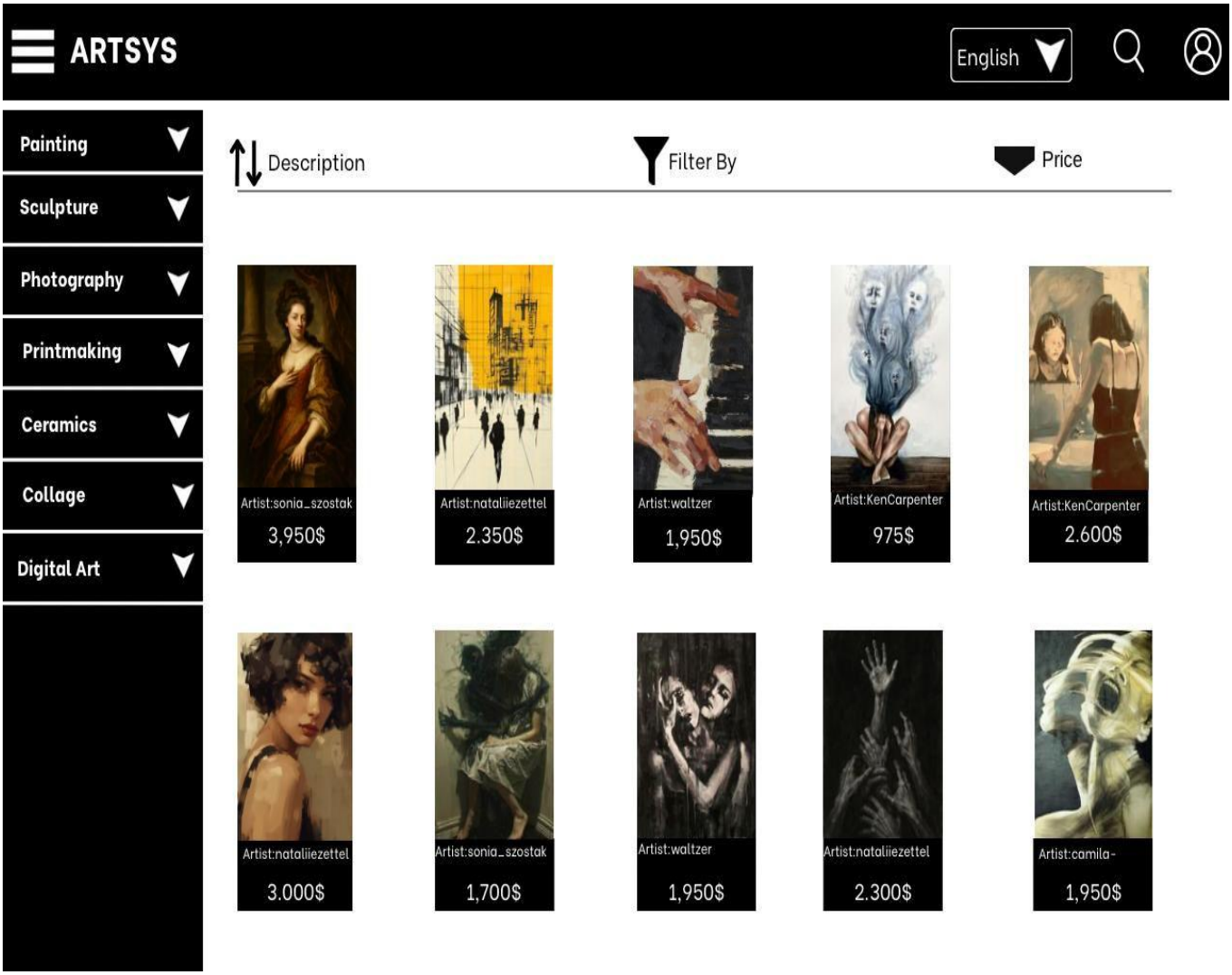
2.Upload Artwork



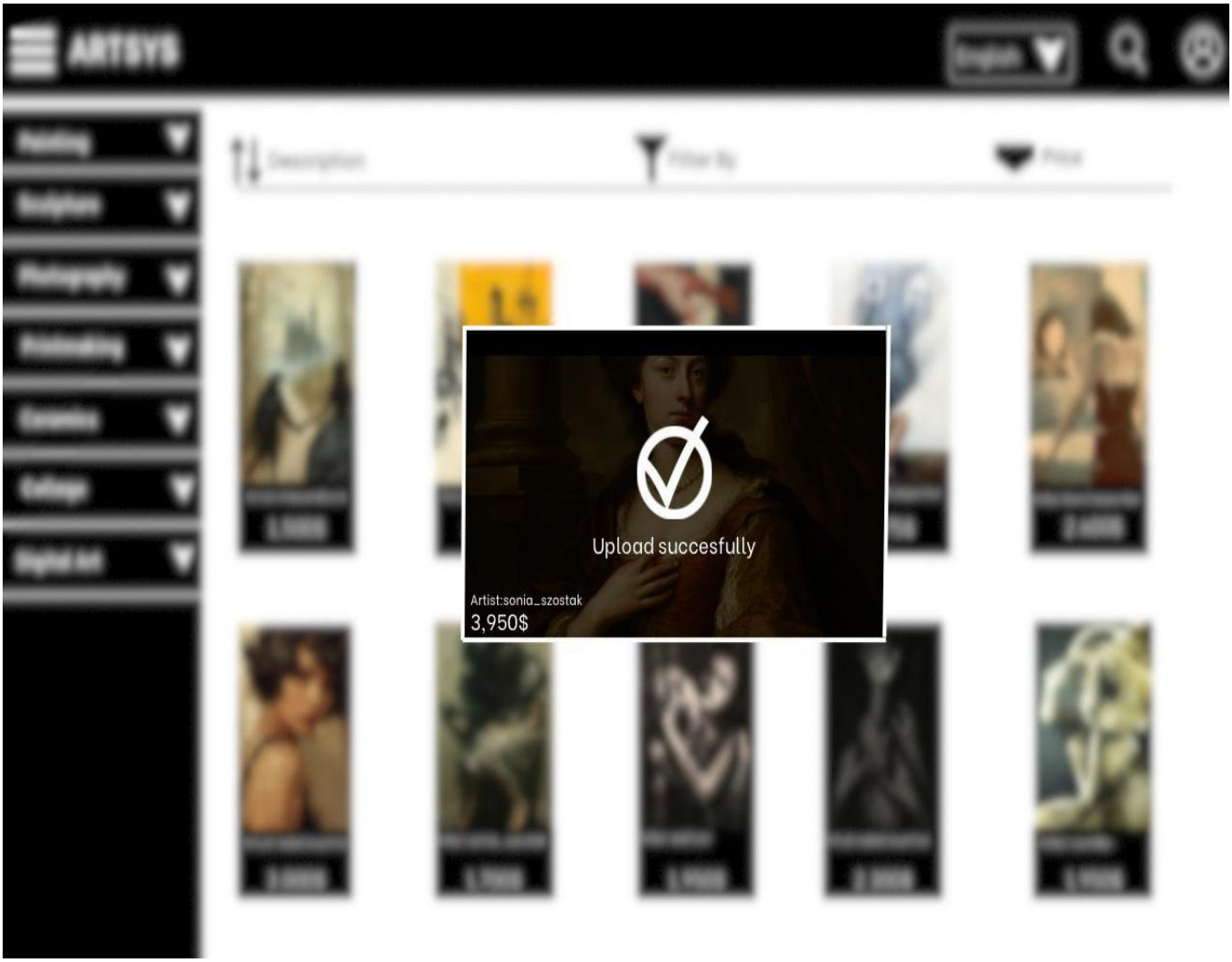
This screenshot displays the initial “Upload New Artwork” page where artists can input artwork metadata including the title, description, category, and price. It also includes a section for image upload. This interface allows the user to start the submission process in an intuitive and user-friendly layout.



This screen shows the artwork upload form filled with sample content. The artist has provided a detailed description and selected a category. An image has also been uploaded and displayed in the preview box. This step demonstrates how the system handles user input validation and display before submission.

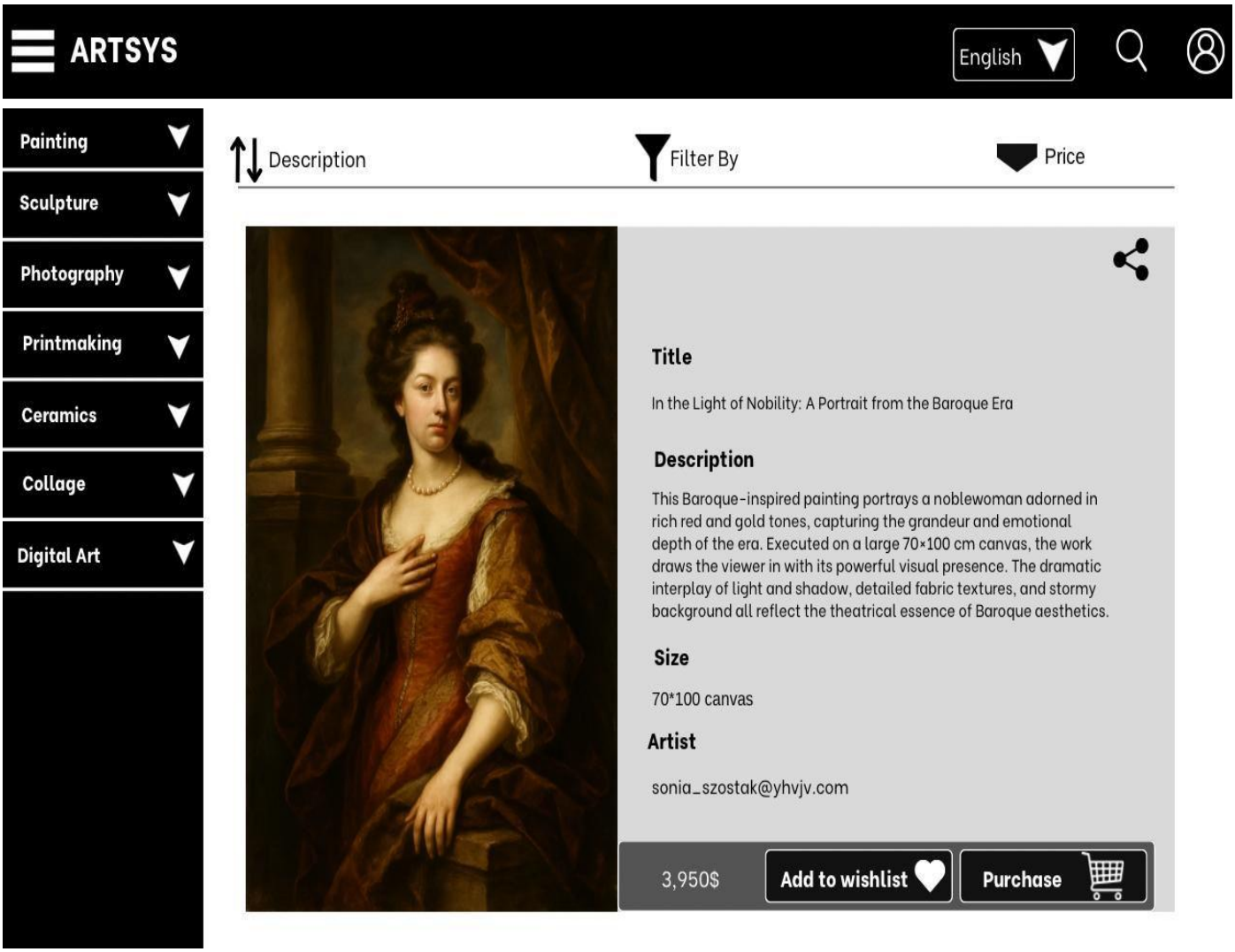


This screenshot represents the artwork being successfully added to the system and displayed in the exhibition listing page. It includes artworks from different artists with filters for category, price, and description, allowing users to browse available items efficiently.



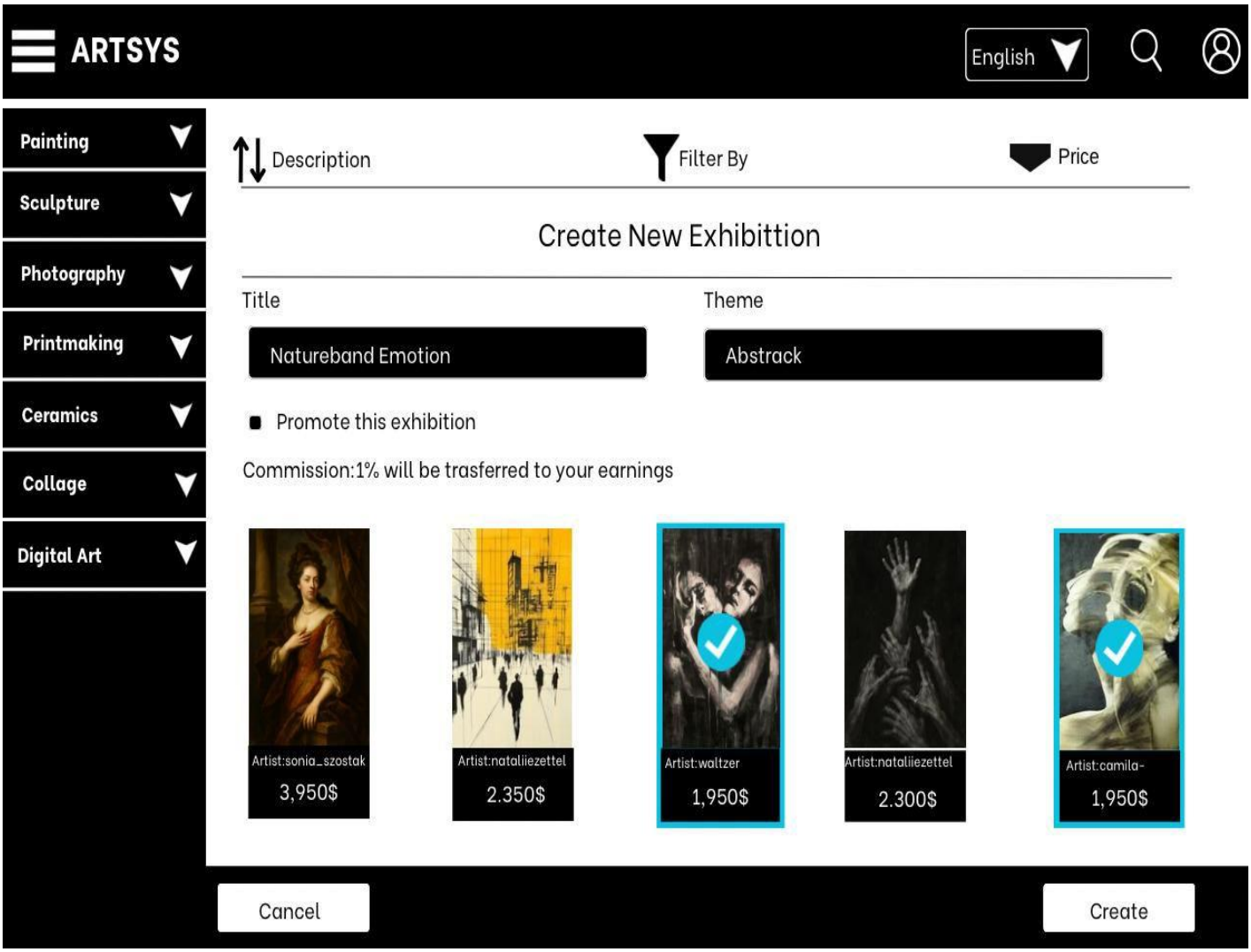
This final confirmation popup appears once the artwork is uploaded successfully. It provides feedback to the artist with the uploaded image and a checkmark icon, enhancing the user experience by confirming successful operation.

3. Purchase Artwork



This detailed view displays a selected artwork with its image, title, description, size, artist email, and price. Users can either add the item to their wishlist or directly purchase it through the provided button. The clean layout ensures that the buyer is well-informed before making a decision.

3. Create Exhibition



This screen enables curators to create a new exhibition by entering a title and theme, selecting artworks from the platform, and optionally promoting the exhibition. Selected artworks are highlighted. The system also informs curators that a small commission will be transferred upon sales from the exhibition.